Honk-Sense

Problem validity, use cases and research findings

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The short answer to our central question. Mirrors, dashcams, parking sensors and radar blind-spot monitors all *look* for objects. None of them *listen*. On Indian roads the horn and the siren are how other road users tell you their intent: "I am here", "I am overtaking", "give way". A sealed, air-conditioned cabin with music on blocks much of that sound, and even when a driver hears it, they often cannot tell where it came from.

Honk-Sense adds value in four places where existing systems cannot help:

1. **Emergency vehicles**: the direction and approach of an ambulance before it is visible.
2. **Sound from out of sight**: a horn around a blind turn or behind a bus or truck. A siren carries where cameras and radar cannot see.
3. **Direction of a horn**: which side a honking two-wheeler or truck is on, in cars that have no radar blind-spot monitoring, which is most cars sold in India today.
4. **Accessibility**: drivers with hearing loss, who have been eligible for licences in India since 2016.

Where existing systems are already enough, Honk-Sense should *not* claim value: parking, silent vehicles in the blind spot (radar's job), and recording evidence (dashcam's job). Our honest position is that Honk-Sense **complements** the existing sensors and does not replace any of them.

How to read this document. [India] marks studies and data from India. † marks a number we found through secondary sources (news or review articles). Check it against the primary report before putting it on a slide. Bracketed codes such as [R1] point to the reference list in Section 12.

1. Objective of the discussion: is the problem valid?

We asked whether the Honk-Sense problem is real and whether it is relevant to Hyundai. The evidence says **yes, with a narrow and honest scope**:

- **Cabins block outside warnings.** A closed car body cuts 21–35 dB at 1 kHz, the band where horns and sirens sit [R12]†. In a US DOT study, drivers with windows closed found the right direction of a siren only **26%** of the time, and sirens were effective only **25–40 ft (8–12 m)** ahead at intersections [R11]†.
- **Music makes it worse.** For every 2 dB of in-car music, the level a siren needs to be heard rises by 1.0–1.7 dB [R13].
- **Drivers miss what they should see.** 75% of lane-change and merge crashes involve a *recognition failure*: the driver looked but did not register the other vehicle [R15].
- **India uses sound as a signal more than most countries.** "Horn OK Please" is painted on trucks, "Blow Horn" signs mark hill turns, and two-wheelers honk to announce themselves. Two IIT/Microsoft India papers built entire traffic-sensing systems on honks for exactly this reason [R6,R7] [India].
- **Delays to ambulances cost lives.** The Law Commission estimated that about **50%** of road deaths could be avoided with care in the first hour [R4]† [India]. A Mumbai survey of ambulance drivers reported that about 65% of private cars do not make way [R5]† [India].

The problem is *not* "drivers cannot see cars"; cameras and radar already address that. The problem is: **important sounds outside the car do not reach the driver in a usable form**, meaning on

time, with a direction, and separated from the noise.

2. Existing systems and their limitations

System	What it solves	Limitations	Where Honk-Sense adds value
Rear-view and side mirrors	Cheap, always there; shows the area behind and to the side	Blind spots remain. Only works if the driver looks: in naturalistic data, 46% looked at the left mirror before a left lane change, and most skipped the over-the-shoulder check [R16,R17]. Poor at night and in rain	A sound cue that does not need the driver to be looking in the right place
Dashcam	Records evidence for insurance and disputes	Records only; gives no live warning. Usually faces forward only	None as a recorder. Honk-Sense is a live warning, not a recording
Ultrasonic parking sensors	Low-speed obstacles while parking	Range of about 2 m, and active only at low speed [R20]	Not relevant for parking; ultrasonic is enough there
Radar blind-spot monitoring (BSM)	Vehicles in the adjacent lane, including silent ones. Cuts lane-change crashes by about 14% [R14]	Available mostly on upper variants in India. Struggles in dense mixed traffic, with two-wheelers weaving and faded lane markings [R18,R19] [India]. Blind to vehicles hidden behind other vehicles	Adds intent (a horn means "I am here" or "I am coming through"). Hears what is behind an obstruction. Lower cost for cars without BSM
Camera ADAS (lane departure, forward collision)	Lanes, vehicles ahead, pedestrians	Needs line of sight. Harder in rain, glare and dust	Hears an ambulance before any camera can see it
Sirens and horns themselves	Warn other road users	Weakened by the cabin, masked by music, and hard to place by direction [R11,R12,R13]	Exactly the gap: brings the warning inside, with its direction

What still goes wrong despite all of these. Every system above is either *visual* (it needs line of sight) or *passive* (it needs the driver to look). None of them carries the *intent* that a horn or siren expresses, and none of them brings that intent into the cabin with a direction.

3. Proposed use cases

We kept the ideas from our discussion, made them sharper, and rated each one honestly. A rating of **Strong** means existing systems clearly cannot do this. **Medium** means Honk-Sense helps, but another system partly covers it. **Supporting only** means it is a secondary benefit, not a headline.

U1. Emergency-vehicle direction and approach — Strong

Situation. An ambulance is 100–200 m behind, hidden by a bus. The driver has music on and windows closed.
Why existing systems fail. Cameras and radar cannot see it. The siren reaches the cabin late, and drivers place it correctly only about a quarter of the time [R11].
What Honk-Sense does. Detects the siren pattern, shows "Ambulance, behind left, approaching",

lowers the music, and suggests which way to move.
Honest limit. The detection range depends on traffic noise, so we should not promise "200 m". Promise *earlier than the driver would notice*, and validate the actual distance in tests.

U2. Direction of a horn: who is honking, and where? Strong

Situation. City traffic, horns from every side. One of them is a bike right at your rear-left quarter as you start to turn left.
Why existing systems fail. The driver hears a horn but cannot tell which one matters or where it is. After hearing hundreds of horns, drivers stop reacting to any of them.
What Honk-Sense does. Shows only the horns that are *close and directed at you*, with an arrow, and ignores the rest. This filtering is the feature.

U3. Blind-spot awareness when the other vehicle honks Medium

Situation. A two-wheeler or truck in your blind spot honks before you change lanes.
Why existing systems fail. Mirrors miss the blind spot, and most cars in India have no radar BSM.
What Honk-Sense does. "Horn, right rear" just as you signal right.
Honest limit. It only works if the other vehicle honks. Radar BSM catches *silent* vehicles too, so say "complements BSM", not "replaces BSM". In Indian traffic, though, vehicles overtaking or filtering usually do honk.

U4. Out of sight: blind turns, ghat roads, behind large vehicles Strong

Situation. A hairpin bend with a "Blow Horn" sign. A bus is coming the other way and honks before it is visible.
Why existing systems fail. Every visual sensor needs line of sight; sound travels around the corner.
What Honk-Sense does. "Vehicle ahead, around the bend", several seconds before it is visible.

U5. Drivers with hearing loss Strong (for this group)

Situation. India has allowed hearing-impaired people to get driving licences since a MoRTH circular in 2016, with further guidance in 2023 [R8][India].
Why existing systems fail. Horns and sirens are audio-only warnings.
What Honk-Sense does. Turns them into visual cues on the cluster and mirror, and optionally a haptic cue in the seat or steering wheel. This is an inclusion story judges will remember.

U6. Fewer interruptions: filtering the noise Medium

Situation. Peak-hour traffic in Hyderabad, where noise often exceeds 90–100 dB [R9,R10][India].
Value. Less stress and less "alarm fatigue". Honk-Sense stays silent for honks not meant for you and speaks up only for the ones that matter.

U7. Level crossings and rural roads Supporting only

Train horns and pressure horns of trucks at night. This is useful, but Indian Railways has removed most unmanned level crossings on its main network, so keep it as a supporting example only.

U8. Fleet and city data — Supporting only

Anonymous counts of where vehicles honk and where sirens get stuck can feed city planning. Nericell [India] showed this was feasible with phones in 2008 [R6]. Treat it as a later-stage idea; it raises privacy questions.

Recommended top five for the pitch: U1, U2, U4, U5 and U3, in that order. U1 and U4 are the strongest because no visual sensor can do them at all.

4. Comparison with existing technologies

Situation	Mirrors	Dashcam	Ultrasonic	Radar BSM	Camera ADAS	Honk-Sense
Ambulance hidden behind traffic	–	–	–	–	–	**Yes**
Horn from behind a blind turn	–	–	–	–	–	**Yes**
Which side a horn came from	–	–	–	–	–	**Yes**
Honking bike in the blind spot	Partly	–	–	Yes	–	**Yes**
Silent car in the blind spot	Partly	–	–	**Yes**	–	No
Parking and low-speed obstacles	Partly	–	**Yes**	–	Partly	No
Lane departure, vehicle ahead	–	–	–	–	**Yes**	No
Evidence after a crash	–	**Yes**	–	–	–	No
Driver with hearing loss	Yes	–	–	Yes	Yes	**Yes**

Where Honk-Sense gives a bigger contribution: anything that is *heard before it is seen*, and anything where the *direction of a sound* matters.
Where existing systems are enough: silent vehicles, parking, lane keeping and evidence. Saying this openly in Q&A builds trust.

5. Research papers and existing studies

Acoustic sensing of honks: Indian work

- **Nericell** (Mohan, Padmanabhan, Ramjee; Microsoft Research India; ACM SenSys 2008) [India]. Phones detected honks and braking on Bengaluru roads, with audio processed on the phone for privacy. It won the ACM Test-of-Time award in 2019 [R6]. *Use:* proves that honk detection in Indian traffic is feasible and that on-device processing is the accepted privacy approach.
- **Horn-OK-Please** (Sen, Raman, Sharma; IIT Bombay; ACM MobiSys 2010) [India]. Estimated vehicle speed from the Doppler shift of honks, using over 18 hours of roadside recordings [R7]. *Use:* shows that honks carry useful motion information, not just presence.

Siren detection and direction finding

- **Listening for Sirens** (Marchegiani and Newman, Oxford, 2018): classifies sirens and horns in city noise and estimates their direction with two microphones [R21].
- **Emergency Vehicles Audio Detection and Localization in Autonomous Driving** (Sun et al., 2021): uses low-cost microphones on a car to estimate siren presence, direction and distance in real driving [R22].
- **SirenNet** (Tran and Tsai, IEEE Access 2020): separates sirens, horns and noise with 98% accuracy on its test set [R23].

- **Acoustic Localization of Emergency Vehicle Sirens** (Ohlenbusch et al., DAGA 2020): an eight-microphone array tested on real traffic [R24].
- **Fraunhofer IDMT "Hearing Car"**, with CARIAD (VW Group): external microphone modules at the rear of the car, AI detection and siren tracking. Tested over 1,500 km in Sweden in 2025 [R25]. *Use:* shows the industry is moving here, while no Indian-traffic product exists.
- **Interior versus exterior microphones** for horns and sirens (Forum Acusticum 2025) [R26]: relevant to where we mount the microphones.
- **Hybrid framework for emergency vehicles on Indian roads** (IEEE 2020, Bengaluru) [India]: camera plus siren pitch detection for traffic-signal priority [R27]. It is infrastructure-side, not in-car, so it leaves our space open.
- **GCC-PHAT** (Knapp and Carter, 1976): the standard method for finding direction from the time difference between microphones.

How audible sirens and horns are inside a car

- US DOT audible-warning study (1970s): effective siren range at an urban intersection of 25–40 ft; drivers with windows closed found the correct direction 26% of the time [R11]†.
- US National Bureau of Standards, 1974: a closed car body removes 21–35 dB at 1 kHz [R12]†.
- Baba and Ebata (Acoustical Society of America, 1996): every 2 dB of music raises the siren detection threshold by 1.0–1.7 dB [R13].
- Cerence driver study (2019): 67% of drivers wanted to be told immediately when a siren is detected and wanted the audio turned down [R28]. This is a survey of preferences, but it supports our "duck the music" feature.

Blind spots and lane changes

- Cicchino (IIHS), *Traffic Injury Prevention* 2018: blind-spot monitoring is linked to **14%** fewer lane-change crashes and 23% fewer with injuries; the injury figure is less statistically certain [R14].
- FMCSA: **75%** of lane-change and merge crashes involve a recognition failure by the driver changing lanes [R15].
- NHTSA 100-Car naturalistic study (DOT HS 811 147, 2009): before a planned left lane change, only 46% of drivers glanced at the left mirror [R16].
- Side blind-zone alert study: drivers skipped the over-the-shoulder glance in 68% of left and 85% of right lane changes. With alerts on, lane changes without a mirror check fell by 31% [R17].

ADAS in Indian conditions

- Kumbhar and Pyasi, SAE 2026-26-0035 [India]: field test of lane-departure warning, AEB and blind-spot monitoring on Indian roads. Effectiveness drops with poor lane markings and unpredictable traffic [R18].
- SAE 2026-26-0042 review and ARAI commentary [India]: unmarked lanes, dense two-wheeler traffic and false alerts are the main challenges for ADAS in India [R19].

Gap we found: no published in-car system detects horns and sirens with direction, trained on Indian traffic sound. Indian work so far is either roadside traffic sensing (Nericell, Horn-OK-Please) or signal-priority infrastructure. This is the clearest evidence that Honk-Sense is new.

6. Indian road conditions and traffic behaviour

Condition	What is different in India	Opportunity for Honk-Sense
Mixed vehicles	Two-wheelers are about 45% of road deaths [R1][India]. Autos, buses, trucks, cycles and carts share one road	Two-wheelers and autos weave close and honk; direction matters most here
Horn as language	"Horn OK Please" on trucks; "Blow Horn" signs on hills [R7][India]	The horn carries intent. Honk-Sense reads that intent
Very loud streets	Traffic noise of 90–100 dB or more measured in Mumbai [R9,R10][India]	Drivers tune out all horns, so filtering only the relevant ones is valuable
Weak lane discipline	Faded or missing markings; vehicles filter between lanes	Camera and radar ADAS lose accuracy [R18][India]; sound does not depend on lane markings
Ghat and hill roads	Hairpin bends with no line of sight	Horn heard around the bend, before the vehicle is visible (U4)
Congested cities	Ambulances stuck in traffic; drivers do not know which way to move [R5][India]	Direction and a suggested move (U1)
Urban and rural highways	Trucks with pressure horns; night driving; music on long trips	Long-trip family drivers with music on are the most cut off
Monsoon and fog	Rain on the roof, glare, low visibility	Visual sensors degrade; sound still travels (noise filtering must handle rain)

7. Accident statistics and supporting data

Data point	Number	Source
Road accidents in India, 2023	4,80,583 accidents; 1,72,890 deaths (about 20 deaths an hour)	MoRTH, *Road Accidents in India 2023* [R1][India]
Two-wheeler riders' share of deaths, 2023	44.8% (about 77,500 deaths)	MoRTH 2023 [R1][India]
Pedestrians' share of deaths, 2023	20.4%	MoRTH 2023 [R1][India]
Hit from back (rear-end)	About 21% of accidents, 20% of deaths	MoRTH 2023 [R2]†[India]
Hit from side	About 15% of accidents	MoRTH 2023 [R2]†[India]
Head-on collisions	About 17% of accidents	MoRTH 2023 [R2]†[India]
Over-speeding, 2022	72.3% of accidents; 71.2% of deaths	MoRTH 2022 [R3]†[India]
Driving on the wrong side, 2022	About 5% of accidents	MoRTH 2022 [R3]†[India]
Deaths avoidable with first-hour care	About 50%	Law Commission, 201st Report [R4]†[India]
Private cars not giving way to ambulances	About 65% (Mumbai, survey of over 100 ambulance drivers)	Radhee Foundation, 2018 [R5]†[India]
Lane-change crashes with recognition failure	75%	FMCSA [R15]
Effect of blind-spot monitoring	14% fewer lane-change crashes	IIHS [R14]

What the data does *not* show, and how to say it.

- MoRTH has **no "blind spot" or "failed to hear" category**. Causes are recorded as speeding,

wrong side and so on. Say so openly: *"India does not record these causes yet; side and rear-end collisions together make up over a third of crashes, and that is where awareness of the surroundings matters."*

- There is **no national figure for ambulance delays caused by traffic**. Use the Law Commission's golden-hour estimate and the Mumbai survey, labelled as one city.
- **Do not claim a percentage reduction for Honk-Sense.** The closest analogy is blind-spot monitoring (14%) and the 31% drop in unchecked lane changes when alerts were on [R14,R17]. Say: "Awareness systems have shown 14–31% improvements in similar situations; we would measure ours in a pilot."

8. Driver psychology and behaviour

- **Habituation and alarm fatigue.** When everything honks, the brain stops responding to horns, including the important ones. Too many alerts cause the same problem with car systems, and drivers switch them off. ARAI studies how false alerts erode trust [R19]. *Design rule:* Honk-Sense must be silent most of the time.
- **Looked but failed to see.** Drivers can look straight at a mirror and still miss a bike; 75% of lane-change crashes involve this kind of recognition failure [R15]. A sound cue with a direction works even when visual attention is elsewhere.
- **Locating sounds inside a car.** People locate sounds well in the open, but a cabin blurs direction, especially front versus back. Only 26% got the siren direction right [R11]. The result is freezing or moving the wrong way when an ambulance comes.
- **Masking by music and conversation.** Music raises the level a siren needs to be heard [R13]. Family trips, the main use case for Hyundai cars, have the most cabin noise.
- **Mirror-checking habits.** Most drivers skip the over-the-shoulder glance [R16,R17]. Two-wheelers filtering between lanes make this more dangerous in India.
- **Stress in congestion.** Constant honking raises stress and irritability, and stressed drivers make more errors. Removing irrelevant honks from the driver's attention is a comfort and safety benefit (U6).
- **Reaction to warnings.** Drivers respond faster to a warning that tells them *what* and *where* than to a general beep. This is why Honk-Sense shows a direction arrow and a label ("Ambulance, behind left"), not just a chime.

9. Installation possibilities

Location	Advantages	Problems
Side-mirror housings (left and right)	Wide spacing between left and right, so good direction. Already has wiring and is fairly sheltered	Wind noise at speed; needs a wind cover
Rear bumper or tailgate	Low wind noise (Fraunhofer's choice [R25]); good for sounds from behind (ambulances, overtaking)	Road splash and dirt; needs sealing
Front bumper or grille	Hears oncoming horns and sounds at blind turns	Engine and fan noise; stone chips
Roof (shark-fin antenna)	360° view; out of splash	Microphones close together, so direction is less precise; wind noise
Inside the cabin	Cheapest; no weather sealing	The cabin blocks and colours outside sound; cabin speech and music get mixed in [R26]
Where the driver sees and hears the alert		
Instrument cluster or HUD	Arrow and label in the line of sight	Must not add clutter
Side-mirror LED	Natural place to look before a lane change	Left and right only
Directional cabin audio	Plays a short cue from the speaker on the correct side; lowers music	Must be brief, not another alarm
Seat or steering vibration	Works for drivers with hearing loss (U5)	Optional; adds cost

Recommendation: four microphones, one at each corner (two in the mirror housings and two at the rear corners), plus an optional front one. This gives direction around the whole car while keeping wind noise manageable. Alerts go to the cluster and mirror LEDs, with a short directional sound cue. This matches the "4 microphones" already in our deck.

10. Why accidents still happen despite existing systems

1. **Mirrors depend on attention.** Drivers do not check them, or check and fail to register [R15,R16].
2. **Blind-spot systems are not universal.** Radar BSM is mostly on upper variants, and in India it is tuned for orderly lanes, not weaving two-wheelers [R18,R19].
3. **Every sensor needs line of sight.** Nothing sees around a bus, a truck or a hairpin bend.
4. **Proximity sensors are short-range and low-speed.** Ultrasonic sensors work up to about 2 m and are made for parking [R20].
5. **Sirens lose to the cabin.** Sealing, AC and music cut them down, and drivers cannot tell where they come from [R11,R12,R13].
6. **Too many horns means no horn.** Habituation makes the meaningful honk invisible.
7. **The gap no one addresses:** *turning outside sound into a short, directional, filtered cue inside the car.* That is Honk-Sense.

11. Expected research outcome and next steps

Outcome we wanted	Status
Validity of the problem statement	Done (Sections 1 and 10)
At least five valuable use cases	Done: 8 use cases, top 5 chosen (Section 3)
Comparison with mirrors, dash-cams, sensors and BSM	Done (Sections 2 and 4)
Research papers, preferably Indian	Done: 2 core Indian papers plus international work (Section 5)
Indian accident statistics	Done, with gaps stated honestly (Section 7)
Driver psychology	Done (Section 8)
Installation locations	Done, with a recommendation (Section 9)

To make it stronger before the final round:

1. Check every † number against the primary report; the MoRTH PDF is the most important.
2. Run a short driver survey of 30–50 people at IIT Hyderabad and in our families. Ask: How often do you hear a horn and not know where it is? When did you last notice an ambulance late? Would you want the music to duck for a siren? This replaces "based on our experience" with real data.
3. Record about 30 minutes of traffic sound from a phone on the dashboard (inside) and outside the window, in Hyderabad traffic and on a highway. Even a simple spectrogram showing how much the cabin blocks sound makes a strong slide.

12. References

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